IGCSE 0607 Practice Questions — E10.3 — Combined Questions & Markscheme

Discrete and continuous data

Total Questions: 12 — **Total Marks:** 54 Marks

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1 A scientist records different types of weather data.
State whether each of the following types of data is discrete or continuous.

(a) The daily maximum temperature.

..... **[1]**

(b) The number of rainy days in a month.

..... **[1]**

(c) The amount of rainfall in one hour.

..... **[1]**

2 A sports analyst collects data during a football match.

(a) State whether the number of goals scored in the match is discrete or continuous.

..... **[1]**

(b) State whether the mass of the football is discrete or continuous.

..... **[1]**

(c) State whether the time taken to complete the match is discrete or continuous.

..... **[1]**

(d) State whether the temperature on the pitch is discrete or continuous.

..... **[1]**

3 A student collects data for a statistics project.
For each of the variables below, state whether the data collected is discrete or continuous.

(a) The number of goals scored in a football match.

..... **[1]**

(b) The mass of a football.

..... **[1]**

(c) The volume of water in a swimming pool.

..... **[1]**

(d) The number of people swimming in a pool.

..... **[1]**

4 Two sets of data are collected during a fitness test.
Data Set A is the exact time, t seconds, taken by athletes to run a race.
Data Set B is the number of push-ups, n, completed by the athletes.

(a) Data Set A is grouped into classes such as $10 \leq t < 12$.
Data Set B is grouped into classes such as $10 - 19$.
Explain why the classes for Data Set A use inequalities while the classes for Data Set B use a hyphen.

.....

..... **[2]**

(b) A student decides to draw a histogram to represent Data Set B. One of the classes is $20 - 29$.

(i) State the lower and upper class boundaries that must be used for this class on the horizontal axis of the histogram.

Lower = ... , Upper = ... **[1]**

(ii) Calculate the class width used for the frequency density calculation.

..... **[1]**

(c) Continuous data can theoretically take any value within a range.
Give an example of a continuous variable that is practically recorded as discrete data in everyday life.

.....

..... **[1]**

5 A student collects data about a local park for a science project.

(a) Explain the mathematical difference between discrete and continuous data.

.....

.....

..... **[2]**

(b) The student measures the lengths of 50 leaves to the nearest millimetre.

(i) State whether the true length of a leaf is discrete or continuous.

..... **[1]**

(ii) Explain why measuring the lengths to the nearest millimetre does not make the variable discrete.

.....

..... **[1]**

6 A farmer weighs 60 apples.
The masses, m grams, are recorded in the table.

Mass (m grams)	**Frequency**
$80 < m \leq 100$	12
$100 < m \leq 120$	26
$120 < m \leq 140$	15
$140 < m \leq 160$	7

(a) State whether the mass of an apple is discrete or continuous data.

..... **[1]**

(b) Calculate an estimate of the mean mass.

..... g **[3]**

(c) Explain why your answer to part (b) is an estimate.

..... **[1]**

7 A scientist measures the lengths of 50 worms and counts the number of spots on each worm.
The frequency table shows the data collected for the number of spots.

Number of spots	$0-4$	$5-9$	$10-14$	$15-19$
Frequency	12	18	14	6

(a) State which of the two variables, length or number of spots, is continuous.

..... **[1]**

(b) The length of one worm is recorded as 14.5 cm, correct to the nearest millimetre.

(i) Write down the lower bound of this length.

..... *cm* **[1]**

(ii) Write down the upper bound of this length.

..... *cm* **[1]**

(c) Using the frequency table, write down the class width for the $5-9$ class.

..... **[2]**

8 Data is collected about the members of an athletics club.

(a) State whether each of the following variables is discrete or continuous.

(i) The number of members in the club.

..... **[1]**

(ii) The exact distance a member jumps in the long jump.

..... **[1]**

(b) The shoe sizes of the members are recorded.
Shoe sizes are available in sizes such as $7, 7.5, 8, 8.5, 9$.
A student claims that because these shoe sizes include decimal values, the data is continuous.
Explain why the student is incorrect.

.....

.....

..... **[2]**

9 A factory packs identical books into boxes.
Each box contains exactly 24 books.

(a) State whether each of the following variables is discrete or continuous.

(i) The number of books in a box.

..... **[1]**

(ii) The mass of a book.

..... **[1]**

(b) The mass of one book is 850 g, correct to the nearest 10 g.
Find the upper bound for the total mass of the 24 books.
Give your answer in kilograms.

..... kg **[2]**

(c) An empty box has a mass of 1.2 kg, correct to 1 decimal place.
Calculate the lower bound for the total mass of a box containing 24 books.

..... kg **[3]**

10 Statistical data can be classified as either discrete or continuous.

(a) State whether each of the following variables is discrete or continuous data.

(i) The number of cars in a car park.

..... **[1]**

(ii) The temperature of a cup of coffee.

..... **[1]**

(iii) The time it takes to complete a jigsaw puzzle.

..... **[1]**

(iv) The number of keys on a piano.

..... **[1]**

11 A botanist studies a new species of plant in a greenhouse.
She records the height, h cm, of each plant and the total number of leaves on each plant.

(a) State whether the height of a plant is discrete or continuous data.

..... **[1]**

(b) The botanist wants to group the heights of the plants into a frequency table.
She suggests using the following class intervals:

$$10 - 19 \qquad 20 - 29 \qquad 30 - 39$$

Explain why these class intervals are not appropriate for the height of the plants.

.....

..... **[2]**

(c) State whether the number of leaves on a plant is discrete or continuous data.

..... **[1]**

12 Measurements and quantities in statistics are defined by their ability to take certain values.

(a) Explain the mathematical difference between continuous data and discrete data.

.....

..... **[2]**

(b) A person's exact foot length is measured in centimetres.
This measurement is then used to determine their shoe size, which can take values such as 7, 7.5, 8, or 8.5.

(i) State whether exact foot length is discrete or continuous data.

..... **[1]**

(ii) A student argues that shoe size must be continuous because it includes decimal values like 7.5.
Explain why the student is incorrect.

.....

.....
[2]

1 **(a)** Continuous **[B1]**

(b) Discrete **[B1]**

(c) Continuous **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Continuous **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Discrete **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Data Set A (time) is continuous, so it can take any decimal value and inequalities are needed to define exact boundary limits without gaps. **[B1]**

Data Set B (number of push-ups) is discrete, so it only takes integer values and a hyphen clearly defines the inclusive start and end integers. **[B1]**

(b) **(i)** 19.5 and 29.5 **[B1]**

(ii) 10 **[B1]**

(c) Age (usually given as an integer in years) OR Money/Currency (rounded to nearest cent/pence) OR Shoe size **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** Discrete data can only take specific or exact values (often counted) **[B1]**

Continuous data can take any value within a given range (often measured) **[B1]**

(b) **(i)** Continuous **[B1]**

(ii) Rounding limits the precision of the recording, but the actual true length could still be any value between the rounded boundaries **[B1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** Continuous **[B1]**

(b) Uses midpoints: 90, 110, 130, 150 **[M1]**

$\Sigma fx = 90 \times 12 + 110 \times 26 + 130 \times 15 + 150 \times 7 = 6940$ **[M1]**

$\frac{6940}{60} = 115.66... = 116$ or 115.7 **[A1]**

(c) The exact masses of the apples in each class are not known (or similar reference to data being grouped) **[B1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** Length **[B1]**

(b) **(i)** 14.45 **[B1]**

(ii) 14.55 **[B1]**

(c) Identifies that the discrete values include $5, 6, 7, 8, 9$ OR uses continuous boundaries $9.5 - 4.5$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(b) The variable can still only take specific, fixed intervals/values (e.g., jumping by 0.5) **[B1]**

It cannot take any value within a range (e.g., a shoe size of 8.23 does not exist) **[B1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(b) Upper bound of mass of one book = 855 g **[M1]**

$24 \times 855 = 20520$ g $\implies$ 20.52 **[A1]**

(c) Lower bound of empty box = 1.15 kg **[B1]**

Lower bound of mass of one book = 845 g = 0.845 kg seen **[B1]**

Total lower bound $= 1.15 + (24 \times 0.845) = 21.43$ **[A1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** Continuous **[B1]**

(b) Height is continuous / can take decimal values. **[B1]**

A plant with a height between the intervals (e.g. 19.5 cm) would not fit into any class. **[B1]**

(c) Discrete **[B1]**

[Back to Question] *[Total Marks: 4]*

12 **(a)** Continuous data can take any value within a given range (measured). **[B1]**

Discrete data can only take specific, exact values (usually counted). **[B1]**

(b) **(i)** Continuous **[B1]**

(ii) Shoe size can only take specific, set values (there are gaps between sizes like 7 and 7.5). **[B1]**

It cannot take every possible value (e.g. you cannot have a shoe size of 7.23), so it must be discrete. **[B1]**

[Back to Question] *[Total Marks: 5]*